Advanced Programming Practice · 21CSC203P

Unit 4 · Session 2 · SLO 1 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 4 : Session – 2 : SLO – 1 | Pure Functions with Examples

I. Match the Following

1. map() — Applies a function to each element. Example: list(map(str.upper, ["a", "b"])) → ["A", "B"].
2. filter() — Selects elements based on a condition. Example: list(filter(lambda x: x > 0, [-1, 2, -3, 4])) → [2, 4].
3. reduce() — Accumulates values to a single result. Example: reduce(lambda a, b: a + b, [1, 2, 3, 4]) → 10.
4. Pure Function — Same output for the same input, with no side effects. Example: def add(a, b): return a + b.

II. Key Notes

• A pure function never mutates its arguments, global variables, or performs I/O — its result depends only on its inputs.
• map() and filter() return lazy iterators in Python 3; wrap them in list() to materialize the results.
• reduce() lives in the functools module: from functools import reduce.